

DIGITAL FORENSICS — PRACTICAL EXAMINATION

Post Graduate Diploma in Cyber Security and Law

Instructions

1. Attempt all 15 questions — each is independent unless stated otherwise.
2. For every question, submit the exact command(s) or GUI steps you used, along with the output/screenshot.
3. Never work on the original evidence file — always duplicate it first and work on the copy.
4. Hash your evidence file before you start and after you finish, and show both hashes.
5. You may use any of the lab tools: dc3dd, dd, Sleuth Kit (fls/icat/mmls/fsstat), exiftool, foremost, FTK Imager, Autopsy, Volatility 3.

Section A — Forensic Imaging & Acquisition

Tools for this section: dd, dc3dd (CLI) · FTK Imager (GUI, Windows) · Autopsy (for viewing the image afterward)

Q1. Create a Forensic Image

Create a 25 MB virtual pendrive image to use as practice evidence for this exam.

- Use dd or dc3dd from the terminal, or use FTK Imager → File → Create Disk Image if working on a Windows lab machine.
- Generate the image's SHA-256 hash immediately after creation and record it — this is your baseline integrity value for the rest of the exam.

Q2. Add a Partition Table and Verify It

Starting from a blank 32 MB image, add a proper MBR partition table with a FAT16/FAT32 partition inside it (instead of a bare filesystem).

- Confirm the partition exists using mmls.
- Confirm the filesystem type and details using fsstat.
- Also open the same image in Autopsy (Add Data Source → Disk Image or VM File) and confirm the Volume/Partition view matches what mmls reported.

Q3. Mount and Browse the Evidence

Mount your image read-only and list every file present on it.

- Do this once via the terminal (mount -o ro,loop).

- Do it a second time by adding the same image as a data source in Autopsy and browsing the file tree in the GUI.
- State one advantage of the GUI view over the terminal view, and one advantage of the terminal view over the GUI.

Section B — Data Recovery & File Carving

Tools for this section: fls, icat (CLI) · Autopsy (Deleted Files / File Carving modules) · foremost

Q4. Recover a Deleted File

On your mounted image, create a file, unmount, delete it (by editing the raw image or using a pre-deleted sample), then recover it.

- Recover it using `fls -d + icat` from the terminal.
- Recover the same file again using Autopsy's "Deleted Files" view under the Data Source tree, and export it from there.
- Confirm both recovered copies produce the same SHA-256 hash as each other.

Q5. Extract Photo Metadata (EXIF/GPS)

You are given a recovered photo, `vacation_photo.jpg`, taken from a suspect's device.

- Extract its metadata using `exiftool` from the terminal — report the Make, Model, Date/Time Original, GPS Latitude/Longitude, and any embedded comment.
- Open the same photo inside Autopsy and locate the equivalent metadata under the file's "Metadata" tab.
- Based on the GPS coordinates, state approximately where the photo was taken.

Q6. Carve Files With No File System

Given a raw disk image with damaged/missing file system metadata, recover any JPEG and PDF files hidden inside it.

- Use `foremost -t jpg,pdf` from the terminal.
- Cross-check your results using Autopsy's built-in file-carving (Unallocated Space analysis) module.
- Briefly explain how carving tools find files without any file system to guide them.

Section C — Formatted Drives & File Systems

Tools for this section: fls, ls -la · xxd · fsstat · Autopsy

Q7. Find Hidden Files on the Evidence Image

A suspect may have hidden files on the drive to avoid casual notice.

- Mount the evidence image and list its contents with a plain ls — note what you see.
- Now reveal hidden/dot-files using ls -la (or list every entry on the raw image with fls -a), or check the same image in Autopsy, which lists hidden files automatically in its file tree.
- Report any hidden files found and explain why an investigator should never trust a “default” file listing alone.

Q8. Identify an Unknown File System

You are handed a raw image with no label and no documentation.

- Use xxd to inspect the boot sector / superblock and identify the magic bytes that reveal the file system type (FAT / NTFS / ext4).
- Confirm your identification using fsstat.

Q9. Timestamp Behaviour Across Media

Create a file on your local machine and note its Created / Modified / Accessed timestamps (stat).

- Copy the file onto a USB drive (or a mounted image) and check its timestamps again.
- State which timestamp(s) changed, which stayed the same, and explain why.

Section D — RAM (Memory) Forensics

Tools for this section: FTK Imager (RAM capture) • Volatility 3 (analysis)

Q10. Capture RAM

On a Windows machine, open 3–5 different applications (e.g. Notepad, a browser, a PDF viewer).

- Use FTK Imager → File → Capture Memory to dump the live RAM to a .mem file.
- State why RAM must be captured before the machine is shut down or the disk is imaged (order of volatility).

Q11. Analyse the Memory Dump — Processes

Using the RAM image captured in Q10 (or a provided sample):

- Run vol -f memdump.mem windows.pslist and windows.pstree.
- Identify each of the applications you opened earlier in the process list output.

Q12. Analyse the Memory Dump — Network Activity

Using the same memory dump:

- Run vol -f memdump.mem windows.netscan.
- Report any active or recently closed network connections and identify the process each one belongs to.

Section E — Log Analysis & Full Investigation

Tools for this section: *grep / awk (CLI) · Autopsy (Timeline module, for Q15)*

Q13. Web Server Attack Detection

Using the sample Apache access.log provided in the lab logs:

- Identify the IP address that is scanning the server for hidden directories/admin panels.
- Report the tool signature (User-Agent) and the time window of the scan.

Q14. SSH Brute-Force Detection

Using the sample auth.log provided in the lab logs:

- Identify the IP address attempting an SSH brute-force attack.
- Justify your answer with the usernames tried and the timestamps involved.

Q15. Complete Investigation (Capstone)

You are given one full evidence image (e.g. the Devid Case image).

- Hash the image, then mount it read-only (terminal) and load it into Autopsy as a new case.
- Identify the file system and partition layout.
- Recover any deleted files (via fls/icat or Autopsy's Deleted Files view).
- Extract EXIF/GPS metadata from any recovered photos (exiftool or Autopsy's Metadata tab).
- Build a simple timeline of file Created/Modified/Accessed times — using mactime, or Autopsy's built-in Timeline feature.
- Write a short (3–4 line) summary of your findings, as you would for a chain-of-custody report.

All 15 questions are drawn from practicals already covered in class. Full worked solutions are provided separately in the Practical Exam Answer Key.